

29. When a beam of light of wavelength λ is incident normally on a plane mirror, it is reflected totally.

The intensity of the light corresponds to a rate of n photons per second.

What is the force exerted on the mirror by the beam of light?

A $nh\lambda$

B $\frac{nh}{\lambda}$

C $2nh\lambda$

D $\frac{2nh}{\lambda}$

Answer: D

momentum of a photon $p = \frac{h}{\lambda}$

change in momentum when photon is reflected completely $= p - (-p) = \frac{2h}{\lambda}$

force exerted = rate of change of momentum of photons $= \frac{\text{no. of photons}}{\text{time}} \times \frac{2h}{\lambda} = \frac{2nh}{\lambda}$